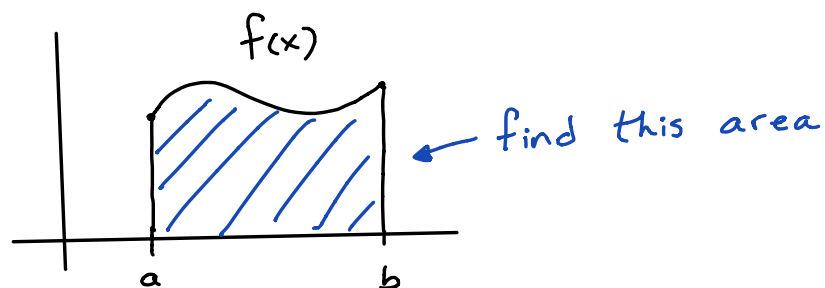


Section 5.2 The Definite Integral



From Section 5.1 we had: $\Delta x = \frac{b-a}{n}$

$$\text{Area} = \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n f(x_i) \cdot \Delta x \right)$$

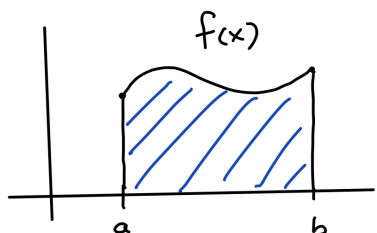
This is a lot to write, so we have notation for this idea.

$$\text{Area} = \int_a^b f(x) \cdot dx$$

$\int f(x) dx$ is called an integral

$\int_a^b f(x) dx$ is called a definite integral

Integral Facts



$$(i) \int_a^a f(x) dx = 0$$

$$(ii) \int_b^a f(x) dx = - \int_a^b f(x) dx$$

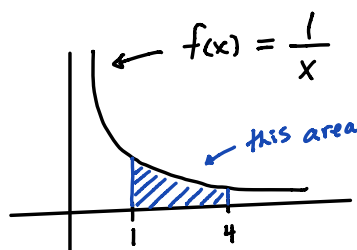
(iii) If k is a constant,

$$\int_a^b k \cdot f(x) dx = k \cdot \int_a^b f(x) dx$$

$$(iv) \int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

Ex ① A) Sketch the area being found by

$$\int_1^4 \frac{1}{x} dx$$

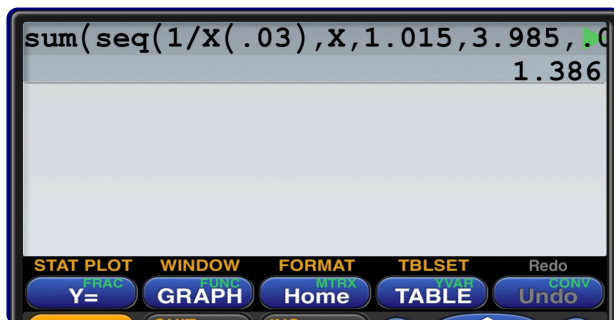


B) Estimate the area using the Midpoints of $n = 100$ rectangles.

$$\Delta x = \frac{4-1}{100} = 0.03$$

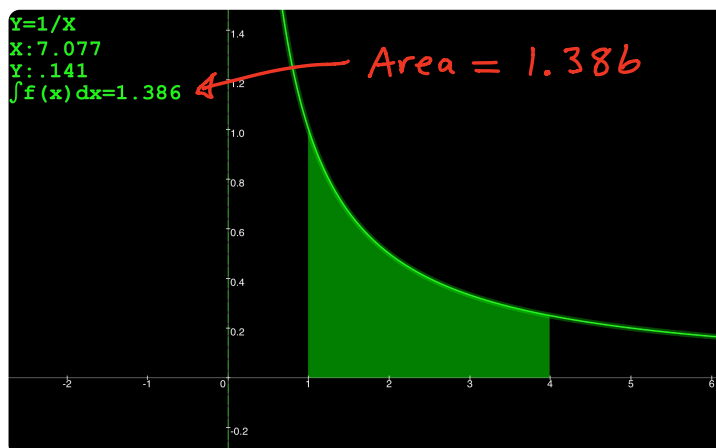


$$\text{Sum}(\text{seq}(\frac{1}{x}(.03), x, 1.015, 3.985, 0.03))$$



Area ≈ 1.386

Let's check this answer in two ways!

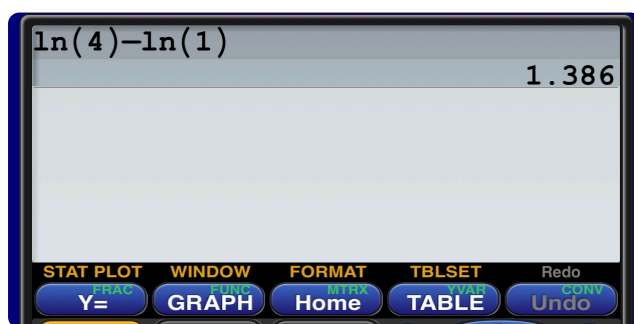


The other way to check the answer is:

$$\text{if } y' = \frac{1}{x}, \text{ then } y = \ln x$$

Since the interval is $[1, 4]$, find

$$\ln 4 - \ln 1$$

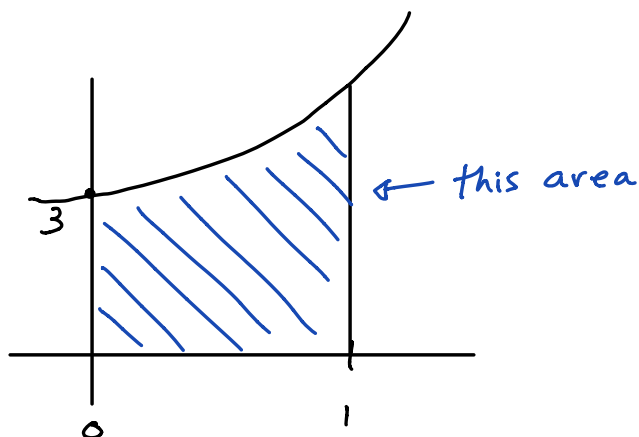


← Area = 1.386

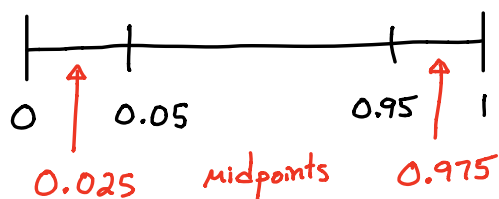
∴

Ex ② $f(x) = e^x + 2$ on $[0, 1]$

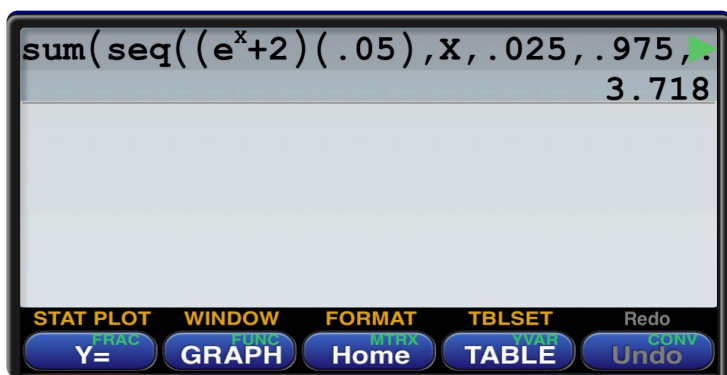
A) Graph the area between $f(x)$ and the x -axis.



B) Use Riemann sums with $n=20$, using the midpoints to estimate the area. $\Delta x = \frac{1-0}{20} = 0.05$

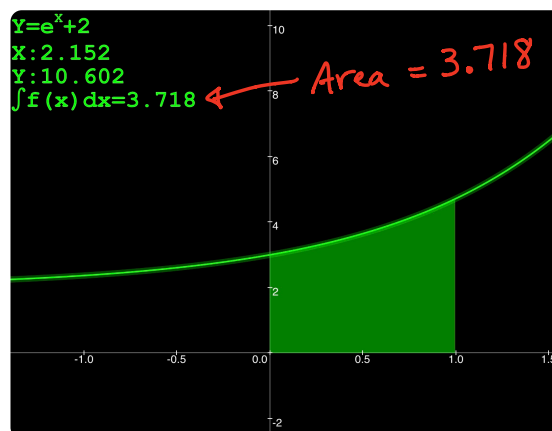


$$\text{Sum}(\text{seq}((e^x+2)(.05), x, 0.025, 0.975, 0.05))$$



← Area ≈ 3.718

Now check the Answer:

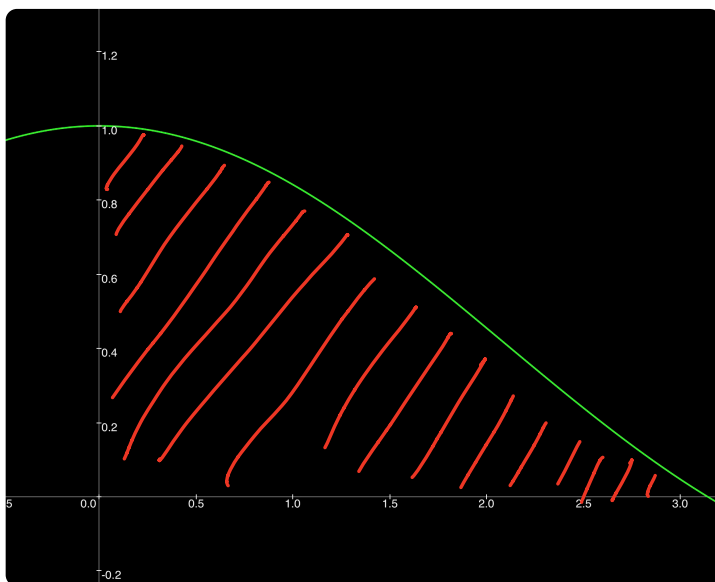


We can also find the exact answer by using the Antiderivative.

$$\begin{aligned}\int_0^1 (e^x + 2) dx &= e^x + 2x \Big|_0^1 \\ &= (e^1 + 2(1)) - (e^0 + 2(0)) \\ &= e + 1 \\ &= 3.718\end{aligned}$$

Ex ③ $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\sin(x_i)}{x_i} \Delta x$ on $[0, \pi]$

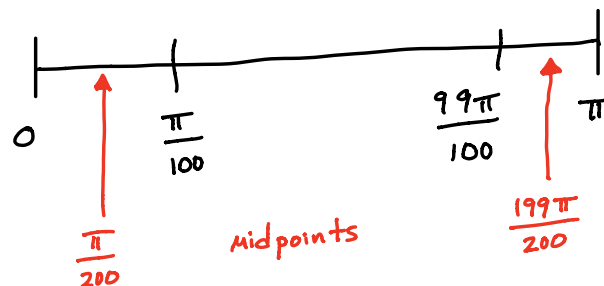
Find the area described above.



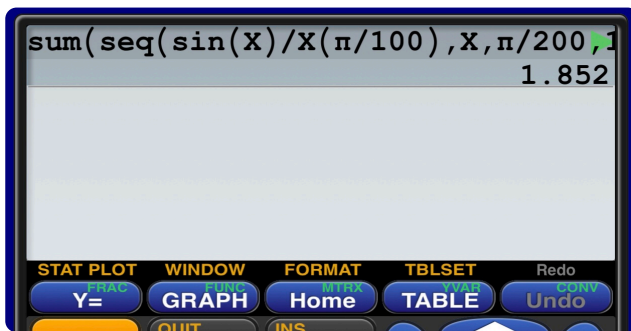
$$f(x) = \frac{\sin x}{x} \text{ on } [0, \pi]$$

Use the midpoints of $n = 100$ rectangles to estimate the area.

$$\Delta x = \frac{\pi - 0}{100} = \frac{\pi}{100}$$



$$\text{Sum} \left(\text{seq} \left(\frac{\sin x}{x} \cdot \frac{\pi}{100}, x, \frac{\pi}{200}, \frac{199\pi}{200}, \frac{\pi}{100} \right) \right)$$



$$\text{Area} \approx 1.852$$

